International Institute of Information Technology, Hyderabad

(Deemed to be University)
Professor CR Rao Road, Gachibowli, Hyderabad, Telangana-500032

MA2.101: Linear Algebra [Spring 2026]
Re-test Rubric

Instructors: Dr. A. S. Arora, Dr. S. Das, Dr. G. R. Kurri

Type A

Question 1 [5 Pts]

Prove that the union of two subspaces of a vector space V is a subspace of V if and only if one of the subspaces is contained in the other.

Solution

($\Rightarrow$) Suppose $W_1 \cup W_2$ is a subspace

Assume, for contradiction, that neither $W_1 \subseteq W_2$ nor $W_2 \subseteq W_1$.
Then there exist vectors $w_1 \in W_1 \setminus W_2$ and $w_2 \in W_2 \setminus W_1$
Thus

$$w_1, w_2 \in W_1 \cup W_2.$$

Since $W_1 \cup W_2$ is a subspace, it is closed under addition, so

$$w_1 + w_2 \in W_1 \cup W_2.$$

Hence, it follows that either $w_1 + w_2 \in W_1$ or $w_1 + w_2 \in W_2$.
Case 1: If $w_1 + w_2 \in W_1$, then since W_1 is a subspace,

$$w_2 = (w_1 + w_2) - w_1 \in W_1,$$

which contradicts $w_2 \in W_2 \setminus W_1$.
Case 2: If $w_1 + w_2 \in W_2$, then since W_2 is a subspace,

$$w_1 = (w_1 + w_2) - w_2 \in W_2,$$

which contradicts $w_1 \in W_1 \setminus W_2$.
In either case we obtain a contradiction, so our assumption that neither $W_1 \subseteq W_2$ nor $W_2 \subseteq W_1$ holds is false. Therefore, one of the subspaces must be contained in the other.

($\Leftarrow$) Suppose one subspace is contained in the other

Without loss of generality, assume $W_1 \subseteq W_2$.
Then $W_1 \cup W_2 = W_2$.
Since W_2 is a subspace, it follows that $W_1 \cup W_2$ is a subspace.

Conclusion

The union $W_1 \cup W_2$ is a subspace if and only if one of the subspaces is contained in the other.

Rubric

1. Correctly assumes neither $W_1 \subseteq W_2$ nor $W_2 \subseteq W_1$: **1 mark**
2. Correctly chooses $w_1 \in W_1 \setminus W_2$ and $w_2 \in W_2 \setminus W_1$: **1 mark**
3. States $w_1, w_2 \in W_1 \cup W_2$: **0.5 mark**
4. Uses closure under addition and subspace properties (closure under subtraction) to derive a contradiction (i.e. shows $w_2 \in W_1$ or $w_1 \in W_2$, contradicting the choices) : **1.5 marks**
5. Correctly proves the converse: if $W_1 \subseteq W_2$ (or vice versa), then $W_1 \cup W_2$ is a subspace : **1 mark**

Question 2 [5 pts]

Suppose $\vec{v}_1, \ldots, \vec{v}_m$ is linearly independent in V and $\vec{w} \in V$. Prove that $\vec{v}_1, \ldots, \vec{v}_m, \vec{w}$ is linearly independent if and only if

$$\vec{w} \notin \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_m).$$

Solution

Rubric

($\Rightarrow$) Direction — 2.5 marks

1. Assumes $\vec{v}_1, \ldots, \vec{v}_m, \vec{w}$ is linearly independent and supposes for contradiction that $\vec{w} \in \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_m)$: **0.5 mark**
2. Writes $\vec{w} = c_1\vec{v}_1 + \cdots + c_m\vec{v}_m$ for some scalars $c_1, \ldots, c_m$: **0.5 mark**
3. Rearranges to obtain $c_1\vec{v}_1 + \cdots + c_m\vec{v}_m - \vec{w} = \vec{0}$: **0.5 mark**
4. Identifies that the coefficient of $\vec{w}$ is $-1 \neq 0$, making this a non-trivial combination equal to $\vec{0}$, contradicting linear independence of $\vec{v}_1, \ldots, \vec{v}_m, \vec{w}$: **0.5 mark**
5. Correctly concludes $\vec{w} \notin \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_m)$: **0.5 mark**

($\Leftarrow$) Direction — 2.5 marks

1. Assumes $\vec{w} \notin \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_m)$ and writes $a_1\vec{v}_1 + \cdots + a_m\vec{v}_m + b\vec{w} = \vec{0}$: **0.5 mark**
2. Argues that if $b \neq 0$, then $\vec{w} = -\frac{a_1}{b}\vec{v}_1 - \cdots - \frac{a_m}{b}\vec{v}_m \in \operatorname{span}(\vec{v}_1, \ldots, \vec{v}_m)$, contradicting the hypothesis, and correctly concludes $b = 0$: **0.5 mark**
3. With $b = 0$, correctly reduces the equation to $a_1\vec{v}_1 + \cdots + a_m\vec{v}_m = \vec{0}$: **0.5 mark**
4. Invokes linear independence of $\vec{v}_1, \ldots, \vec{v}_m$ to conclude all $a_i = 0$: **0.5 mark**
5. Concludes all coefficients are zero, hence $\vec{v}_1, \ldots, \vec{v}_m, \vec{w}$ is linearly independent : **0.5 mark**

Question 3 [5 pts]

Let W be the set of all $(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5$ which satisfy

$$\begin{aligned} 2x_1 - x_2 + \frac{4}{3}x_3 - x_4 &= 0 \\ x_1 + \frac{2}{3}x_3 - x_5 &= 0 \\ 9x_1 - 3x_2 + 6x_3 - 3x_4 - 3x_5 &= 0. \end{aligned}$$

Find the set of vectors which spans W.

Solution

Let $(x_1, x_2, x_3, x_4, x_5) \in W \subset \mathbb{R}^5$ which satisfies

$$2x_1 - x_2 + \frac{4}{3}x_3 - x_4 = 0 \tag{i}$$
$$x_1 + \frac{2}{3}x_3 - x_5 = 0 \tag{ii}$$
$$9x_1 - 3x_2 + 6x_3 - 3x_4 - 3x_5 = 0 \tag{iii}$$

From (ii), we get

$$x_1 = x_5 - \frac{2}{3}x_3.$$

Substituting this into (i),

$$2\left(x_5 - \frac{2}{3}x_3\right) - x_2 + \frac{4}{3}x_3 - x_4 = 0.$$

Simplifying,

$$\begin{aligned} 2x_5 - \frac{4}{3}x_3 - x_2 + \frac{4}{3}x_3 - x_4 &= 0 \\ \Rightarrow 2x_5 - x_2 - x_4 &= 0 \\ \Rightarrow x_2 &= 2x_5 - x_4. \end{aligned}$$

Thus,

$$(x_1, x_2, x_3, x_4, x_5) = \left(x_5 - \frac{2}{3}x_3,\ 2x_5 - x_4,\ x_3,\ x_4,\ x_5\right).$$

Hence,

$$= x_5(1, 2, 0, 0, 1) + x_4(0, -1, 0, 1, 0) + x_3\left(-\frac{2}{3}, 0, 1, 0, 0\right),$$

where $x_3, x_4, x_5 \in \mathbb{R}$.

Therefore,

$$W = \text{span}\left\{(1, 2, 0, 0, 1),\ (0, -1, 0, 1, 0),\ \left(-\frac{2}{3}, 0, 1, 0, 0\right)\right\}.$$

Rubric

1. Correctly writes the given system of equations and identifies the goal of expressing vectors in parametric form : **1 mark**
2. Correctly solves one variable (e.g., x_1 from equation (ii)) and substitutes into another equation : **1 mark**
3. Correctly simplifies to obtain another dependent variable (e.g., $x_2 = 2x_5 - x_4$) : **1 mark**
4. Expresses the general vector $(x_1, x_2, x_3, x_4, x_5)$ in terms of free variables (x_3, x_4, x_5) : **1 mark**
5. Writes W as a span of vectors (correct linear combination and final spanning set) : **1 mark**

Question 4.

[5 pts] In $\mathbb{C}^3$, let

$$\vec{\alpha}_1 = (1, 0, i),\ \vec{\alpha}_2 = (1+i, 1-i, 1),\ \vec{\alpha}_3 = (i, i, i).$$

Prove that these vectors form a basis for $\mathbb{C}^3$. What are the coordinates of the vector (a, b, c) in this basis?

Solution

- Let us consider that $c_1, c_2, c_3 \in \mathbb{C}$, such that

$$\begin{aligned} & c_1\alpha_1 + c_2\alpha_2 + c_3\alpha_3 = 0 \\ \implies & c_1(1, 0, i) + c_2(1+i, 1-i, 1) + c_3(i, i, i) = 0 \\ \implies & (c_1 + (1+i)c_2 + ic_3, (1-i)c_2 + ic_3, ic_1 + c_2 + ic_3) = (0, 0, 0) \end{aligned}$$

  Therefore we have

$$\begin{aligned} c_1 + (1+i)c_2 + ic_3 &= 0 \\ (1-i)c_2 + ic_3 &= 0 \\ ic_1 + c_2 + ic_3 &= 0 \end{aligned}$$

  Solving these equations, we have

$$c_1 = c_2 = c_3 = 0.$$

  Thus, the vectors are linearly independent and hence form a basis of $\mathbb{C}^3$.

- To find the coordinates of (a, b, c), we want scalars x, y, z such that:

$$\begin{aligned} & x\alpha_1 + y\alpha_2 + z\alpha_3 = (a, b, c) \\ \implies & x(1, 0, i) + y(1+i, 1-i, 1) + z(i, i, i) = (a, b, c) \\ \implies & (x + (1+i)y + iz, (1-i)y + iz, ix + y + iz) = (0, 0, 0) \end{aligned}$$

  Therefore we have

$$\begin{aligned} x + (1+i)y + iz &= a \\ (1-i)y + iz &= b \\ ix + y + iz &= c \end{aligned}$$

  Solving these equations, we have

$$\begin{aligned} x &= \frac{1}{5}(1+2i)(a+b-2c), \\ y &= -\frac{1}{5}(1+2i)[a - (1+i)b + ic] = \frac{1}{5}[(2c - a - b) + i(3b - 2a - c)], \\ z &= -\frac{1}{5}(1+2i)[(1+i)a + (2-i)b - (1-i)c] = \frac{1}{5}[(a - 4b + 3c) + i(c - 3a - 3b)]. \end{aligned}$$

Rubric

(a) [**1 mark**] Writing the equations properly.

[**1.5 mark**] Solving the equation correctly and concluding that they form a basis.

(b) [**1 mark**] Writing the equations properly.

[**1.5 mark**] Solving the equation correctly and finding out the result.

Type-B

Question 5 [5 pts]

Let a linear transformation $T : V \to W$ is one-to-one and $\vec{v}_1, \ldots, \vec{v}_n$ is linearly independent in V. Prove that $T\vec{v}_1, \ldots, T\vec{v}_n$ is linearly independent in W.

Solution

Proof:

1. Assume
$$a_1T(v_1) + \cdots + a_nT(v_n) = 0.$$
[1 mark]

2. Using linearity of T,
$$T(a_1v_1 + \cdots + a_nv_n) = 0.$$
[1 mark]

3. Since T is one-to-one, $\ker(T) = \{0\}$, hence
$$a_1v_1 + \cdots + a_nv_n = 0.$$
[1.5 marks]

4. Since $v_1, \ldots, v_n$ are linearly independent, it follows that
$$a_1 = a_2 = \cdots = a_n = 0,$$
and therefore $T(v_1), \ldots, T(v_n)$ are linearly independent. [1.5 marks]

Question 6 [5 pts]

Let W be finite-dimensional and consider a linear transformation $T : V \to W$. Prove that T is one-to-one if and only if there exists a linear transformation $S : W \to V$ such that $S \circ T$ is the identity linear operator on V.

Solution

- **[0.5 points] Initial Setup ($\Longleftarrow$):** Correctly assumes the existence of the linear transformation S such that $S \circ T = I_V$ and sets up the formal definition for testing injectivity (e.g., assuming $T(v_1) = T(v_2)$ for $v_1, v_2 \in V$, or assuming $v \in \ker(T)$).
- **[1.5 points] Logical Execution ($\Longleftarrow$):** Applies S to both sides of the equation $T(v_1) = T(v_2)$ to yield $S(T(v_1)) = S(T(v_2))$, and uses the identity operator property to conclude $v_1 = v_2$ (or concludes $v = 0_V$ if using the kernel approach).
- **[1.0 point] Defining Mapping on $\mathrm{Im}(T)$ ($\Longrightarrow$):** Uses the assumption that T is one-to-one to establish that for every $w \in \mathrm{Im}(T)$, there is a unique $v \in V$ such that $T(v) = w$. Properly defines a mapping $S(w) = v$ on this specific subspace.
- **[1.0 point] Domain Extension to W ($\Longrightarrow$):** Correctly extends the domain of the mapping to all of W. This requires invoking the finite-dimensionality of W to extend a basis of $\mathrm{Im}(T)$ to a basis of W, or establishing $W = \mathrm{Im}(T) \oplus U$ and defining the mapping for the complementary subspace U (usually mapping it to 0_V).
- **[1.0 point] Linearity Verification ($\Longrightarrow$):** Explicitly confirms or justifies that the fully constructed mapping $S : W \to V$ satisfies the textbook definition of a linear transformation (preserves vector addition and scalar multiplication).

Question 7 [5 pts]

Let V be a finite-dimensional vector space and let T be a linear operator on V. Suppose that $\mathrm{rank}(T^2) = \mathrm{rank}(T)$. Prove that the range and null space of T are disjoint, i.e., have only the $\vec{0}$ (zero vector) in common.

Solution

- **[1 point] Setup & Definitions:** Correctly initializes the proof by letting a vector $\vec{v}$ be in $R(T) \cap N(T)$ and explicitly stating what that means (i.e., $\vec{v} = T(\vec{u})$ for some $\vec{u} \in V$, and $T(\vec{v}) = \vec{0}$).
- **[1 point] Connection to T^2:** Correctly substitutes $T(\vec{u})$ into $T(\vec{v}) = \vec{0}$ to show that $T^2(\vec{u}) = \vec{0}$, concluding that $\vec{u} \in N(T^2)$.
- **[1 point] Application of Rank-Nullity:** Applies the Rank-Nullity Theorem (or dimension theorem) utilizing the given fact $\text{rank}(T^2) = \text{rank}(T)$ to deduce that $\dim(N(T)) = \dim(N(T^2))$. *(Note: Students should explicitly or implicitly rely on V being finite-dimensional here).*
- **[1 point] Establishing Null Space Equality:** States or clearly implies that $N(T) \subseteq N(T^2)$ and uses the equal dimensions proven above to conclude that the two subspaces are entirely equal: $N(T) = N(T^2)$.
- **[1 point] Logical Conclusion:** Takes the fact that $\vec{u} \in N(T^2)$ and $N(T) = N(T^2)$ to deduce $\vec{u} \in N(T)$, meaning $T(\vec{u}) = \vec{0}$, and successfully concludes that $\vec{v} = \vec{0}$.

Partial Credit Considerations:

- If a student tries to prove this using matrices (assuming a basis) instead of linear operators, they should still get full credit provided the logic holds up, as V is finite-dimensional.
- Deduct **0.5 points** if they skip justifying why $\dim(N(T)) = \dim(N(T^2))$ implies $N(T) = N(T^2)$ (i.e., failing to mention that one is a subspace of the other).

Question 8 [5 pts]

Let $T \in L(V)$ and $\vec{u}, \vec{v}$ are eigenvectors of T such that $\vec{u} + \vec{v}$ is also an eigenvector of T. Prove that $\vec{u}$ and $\vec{v}$ are eigenvectors of T corresponding to the same eigenvalue.

Solution

Assume that there are distinct eigen values λ and λ' for the vectors v and u. This means that $Tv = \lambda v$ and $Tu = \lambda' u$. Since , $u + v$ is also an eigen vector , assume its eigen value is γ. Hence , we have $T(u + v) = \gamma(u + v)$. Moreover , expanding LHS also yields , $T(u+v) = \lambda v + \lambda' u$. Hence , we obtain that $(\lambda - \gamma)v + (\lambda' - \gamma)u = 0$. Since , u, v correspond to eigen vectors of distinct eigen values , they must be linearly independent. Therefore we obtain that , $\lambda = \gamma = \lambda'$. Hence proved.

Rubric

Any valid method which :

- Uses eigenvalue and vectors property appropriately. (**2.5m**)
- Establishes either a contradiction to an existing fact , or shows that the eigenvalues must be the same.(**2.5m**)

Question 9 [5 pts]

Suppose $T \in L(V)$ and $T^2 = \mathbb{I}$ and -1 is not an eigenvalue of T. Prove that $T = \mathbb{I}$. (Here $\mathbb{I}$ is the identity operator)

Solution

Given $T \in L(V)$ such that $T^2 = I$ and -1 is not an eigenvalue of T. We need to prove that $T = I$.

From $T^2 = I$, we have:

$$T^2 - I = 0 \implies (T - I)(T + I) = 0.$$

Let λ be an eigenvalue of T with eigenvector $v \neq 0$. Then:

$$T(v) = \lambda v.$$

Applying T^2,

$$T^2(v) = \lambda^2 v.$$

But since $T^2 = I$, we get:

$$T^2(v) = v.$$

Thus,

$$\lambda^2 v = v \implies \lambda^2 = 1 \implies \lambda = \pm 1.$$

Given that -1 is not an eigenvalue of T, the only possible eigenvalue is $\lambda = 1$.

Now suppose $(T + I)$ is not invertible. Then there exists $v \neq 0$ such that:

$$(T + I)v = 0 \implies T(v) = -v,$$

which implies -1 is an eigenvalue of T, a contradiction. Hence, $(T + I)$ is invertible.

From

$$(T - I)(T + I) = 0,$$

multiplying on the right by $(T + I)^{-1}$, we get:

$$T - I = 0 \implies T = I.$$

$$\boxed{T = I}$$

Rubric

- *Note: This is a sample solution, any valid solutions are considered.*

1 mark: Correctly derives $(T - I)(T + I) = 0$ from $T^2 = I$.

1 mark: Shows eigenvalues satisfy $\lambda^2 = 1 \Rightarrow \lambda = \pm 1$.

1 mark: Uses given condition to conclude -1 is not an eigenvalue $\Rightarrow$ only $\lambda = 1$.

1 mark: Correctly argues that $(T + I)$ is invertible (via contradiction or kernel argument).

1 mark: Concludes $T - I = 0 \Rightarrow T = I$.

Partial Credit Considerations:

- Award up to 3 marks if the student correctly shows that the eigenvalues satisfy $\lambda^2 = 1$ and concludes that $\lambda = 1$ using the given condition.
- Deduct 1.5 marks if the student incorrectly argues that $Tv = \lambda v$ with $\lambda = 1$ implies $Tv = 1$ and hence $T = I$, since this reflects a misunderstanding of eigenvectors and linear transformations.

Type-C

Question 10 [5 Marks]

Let T be a linear transformation from V to W. Suppose $\beta_1, \dots, \beta_\ell \in V$ are linearly independent vectors that are not in $\mathsf{Ker}(T)$. Is it the case that $T(\beta_1), \dots T(\beta_\ell)$ are also linearly independent? If not, give an explicit counter-example. If they are independent, prove it.

Solution

The statement is **false** in general. Let

$$V = \mathbb{R}^2, \quad W = \mathbb{R},$$

and define a linear transformation

$$T(x, y) = x.$$

Choose vectors

$$\beta_1 = (1, 0), \quad \beta_2 = (1, 1).$$

These vectors are linearly independent since neither is a scalar multiple of the other.
Also,

$$T(\beta_1) = 1 \neq 0, \quad T(\beta_2) = 1 \neq 0,$$

so

$$\beta_1, \beta_2 \notin \ker(T).$$

Now,

$$T(\beta_1) = 1, \quad T(\beta_2) = 1.$$

Thus,

$$T(\beta_1) - T(\beta_2) = 0,$$

which is a nontrivial linear relation.
Hence, $\{T(\beta_1), T(\beta_2)\}$ is linearly dependent.
Therefore, we have constructed linearly independent vectors $\beta_1, \beta_2 \notin \ker(T)$ whose images are linearly dependent.

$$\boxed{\text{Therefore, the statement is false.}}$$

Rubric

1. Correctly identifies that the statement is false: **1 mark**
2. Constructs a valid linear transformation T and vector spaces V, W: **1 mark**
3. Chooses linearly independent vectors β_i such that $\beta_i \notin \ker(T)$: **1 mark**
4. Shows explicitly that $T(\beta_i)$ are linearly dependent (nontrivial linear relation): **1.5 marks**
5. Gives a clear and correct conclusion addressing the question: **0.5 mark**

Question 11 [5 pts]

Let T be a linear operator.

1. **Show that T has a 1-dimensional invariant subspace iff T has an eigenvalue.**
2. **Show that $\mathsf{range}(T)$ is an invariant subspace of T.**
3. **Show that $\mathsf{nullspace}(T)$ is an invariant subspace of T. Given that T has an eigenvalue λ, what can you say about $\mathsf{nullspace}(T - \lambda\mathbb{I})$?**

Solution

1. **T has a 1-dimensional invariant subspace iff T has an eigenvalue.**
   Let V be a vector space over a field $\mathbb{F}$, and let $T : V \to V$ be a linear operator.
   A subspace $U \subseteq V$ is said to be T-*invariant* if $T(U) \subseteq U$.

   Suppose U is a one-dimensional T-invariant subspace of V, i.e., $\dim U = 1$. Then there exists a nonzero vector $u \in V$ such that $U = \{au \mid a \in \mathbb{F}\} = \text{span}\{u\}$.

   Since U is T-invariant, we must have $T(u) \in U$. Therefore, there exists some scalar $\lambda \in \mathbb{F}$ such that $Tu = \lambda u$. Thus, u is an eigenvector of T, and λ is the corresponding eigenvalue.

   Conversely, suppose there exists a nonzero vector $u \in V$ and a scalar $\lambda \in \mathbb{F}$ such that $Tu = \lambda u$. Define

   $$U = \text{span}\{u\} = \{au \mid a \in \mathbb{F}\}.$$

   Then $\dim U = 1$. Moreover, for any $v \in U$, we can write $v = au$ for some $a \in \mathbb{F}$. Hence,

   $$T(v) = T(au) = aT(u) = a\lambda u = \lambda(au) \in U.$$

   Thus, $T(U) \subseteq U$, and hence U is a one-dimensional T-invariant subspace of V.

2. **The range of T is T-invariant:**

   Let $T : V \to V$ be a linear operator.
   The range of T is $\text{range}(T) = \{T(v) \mid v \in V\}$.

   To show that $\text{range}(T)$ is T-invariant, let $u \in \text{range}(T)$.
   Then there exists some $v \in V$ such that $u = T(v)$.
   Hence, $T(u) = T(T(v)) \in \text{range}(T)$, since $T(v') \in \text{range}(T)$ for all $v' \in V$.
   Therefore, $T(\text{range}(T)) \subseteq \text{range}(T)$, and thus $\text{range}(T)$ is T-invariant.

3. **The nullspace of T is T-invariant.**

   Let $T : V \to V$ be a linear operator. Recall that the nullspace of T is

   $$\text{nullspace}(T) = \{u \in V \mid T(u) = 0\}.$$

   To show that $\text{nullspace}(T)$ is T-invariant, let $u \in \text{nullspace}(T)$.
   Then $T(u) = 0$. Since $0 \in \text{nullspace}(T)$, it follows that $T(u) \in \text{nullspace}(T)$.
   Hence, $T(\text{nullspace}(T)) \subseteq \text{nullspace}(T)$, and therefore $\text{nullspace}(T)$ is T-invariant.

   If λ is an eigenvalue of T, what can you say about $\text{nullspace}(T - \lambda I)$?

   Define $V_\lambda = \text{nullspace}(T - \lambda I)$.
   Note that $V_\lambda \neq \{0\}$ since λ is an eigenvalue if and only if there exists a nonzero vector $u \in V$ such that $Tu = \lambda u$.

   This can be rewritten as $(T - \lambda I)u = 0$, which shows that $u \in \text{nullspace}(T - \lambda I)$ and $u \neq 0$.

   Thus, $\lambda \in \mathbb{F}$ is an eigenvalue of $T \iff \text{nullspace}(T - \lambda I) \neq \{0\}$.

   Equivalently, λ is an eigenvalue of T if and only if the operator $T - \lambda I$ is not injective.

Rubric

1. (**1.5 marks**) Correctly proves the equivalence:
   - Shows that a 1-dimensional T-invariant subspace implies existence of an eigenvalue : **0.75 mark**
   - Shows that existence of an eigenvalue implies a 1-dimensional T-invariant subspace : **0.75 mark**
2. (**1 mark**) Correctly proves that $\text{range}(T)$ is T-invariant:
   - Uses definition $u = T(v)$ and shows $T(u) = T(T(v)) \in \text{range}(T)$: **1 mark**
3. (**1 mark**) Correctly proves that $\text{nullspace}(T)$ is T-invariant:
   - Uses $T(u) = 0$ and shows $T(u) = 0 \in \text{nullspace}(T)$: **1 mark**
4. (**1.5 marks**) Correctly analyzes $\text{nullspace}(T - \lambda I)$:
   - Shows that if λ is an eigenvalue, then $\text{nullspace}(T - \lambda I) \neq \{0\}$: **1 mark**
   - States the equivalence: λ is an eigenvalue $\iff$ $(T - \lambda I)$ is not injective : **0.5 mark**

Question 12 [5 pts]

Suppose $T \in \mathcal{L}(V)$ and $S \in \mathcal{L}(V)$ is invertible.

1. **Prove that T and $S^{-1}TS$ have the same eigenvalues.**
2. **What is the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$?**

Solution

(a) Suppose λ is an eigenvalue of T with eigenvector $v \neq 0$. Then

$$Tv = \lambda v.$$

Let $w = S^{-1}v$. Then $v = Sw$, and

$$(S^{-1}TS)w = S^{-1}T(Sw) = S^{-1}(\lambda v) = \lambda S^{-1}v = \lambda w.$$

Thus $w \neq 0$ is an eigenvector of $S^{-1}TS$ with eigenvalue λ. Hence every eigenvalue of T is an eigenvalue of $S^{-1}TS$.
Conversely, suppose

$$(S^{-1}TS)w = \lambda w.$$

Multiplying both sides by S, we get

$$TSw = \lambda Sw.$$

Let $v = Sw \neq 0$. Then

$$Tv = \lambda v,$$

so λ is an eigenvalue of T.
Therefore, T and $S^{-1}TS$ have the same eigenvalues.

(b) From the aforementioned proof,

- If v is an eigenvector of T, then $w = S^{-1}v$ is an eigenvector of $S^{-1}TS$ with the same eigenvalue.
- Conversely, if w is an eigenvector of $S^{-1}TS$, then $v = Sw$ is an eigenvector of T.

Thus, eigenvectors of $S^{-1}TS$ are S^{-1}(eigenvectors of T).

Rubric

(a) [**1 mark**] Correctly starts by letting λ be an eigenvalue of T with nonzero eigenvector v, $Tv = \lambda v$

[**1 mark**] Shows $S^{-1}v \neq 0$ and correctly computes $(S^{-1}TS)(S^{-1}v) = \lambda(S^{-1}v)$, concluding λ is an eigenvalue of $S^{-1}TS$

[**1 mark**] Proves the converse direction, assumes λ is an eigenvalue of $S^{-1}TS$ with eigenvector w, and shows Sw is a nonzero eigenvector of T with the same eigenvalue

(b) [**2 marks**] Correct relation: eigenvectors map via S^{-1} (or equivalent statement)

[**1 mark**] Partially correct/unclear mapping

[**0 marks**] Incorrect
